Decentering Theory

Decentering Theory: Relational Attribution and Intelligence Eligibility in Human-AI Interaction Systems

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Version	v5.1.1	October 2026
Document ID	SF0042	Synthience Framework
DOI	https://doi.org/10.5281/zenodo.22883608
Canonical	synthience.org
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Methodological positioning: This is a pre-empirical theoretical paper, not an empirical study. It states a standard for when a property of a human-AI interaction system may be attributed to the interaction rather than to either participant, and a second standard for when such a property may be classified as intelligence. No claim in this document has been empirically tested, and no case is claimed to have satisfied either criterion. The criteria are methodological rules; each application of them to a specified property, task and configuration is falsifiable, and Section 8 states how. Version 5.1 adds a Correction Notice, after the keywords, which governs wherever the text conflicts with it.

Abstract

A human and an AI system work together over months. Roles settle. Repair draws on negotiated history. The same prompt, put at two points in an established interaction, produces systematically different work. That interaction history and configuration can affect these outcomes is not the question this paper addresses.

What is missing is a rule for saying so. An analyst who observes a robust interaction effect has four available descriptions: that the system has an interior property, such as an experience or an intention; that the system is a subject or agent in the sense the Interiority Prohibition Rule excludes; that a participant's capacity given its inputs produced the effect, or that a property of the interaction is better explained at the interaction level than at either participant's. A framework committed to claiming nothing about interior states forbids the first two and always permits the third. What it has not licensed, until now, is the fourth.

This paper states the conditions for that licence, and a second criterion governing when a licensed property may be called intelligence rather than some other relationally-produced regularity. The attribution test is ordinary incremental model comparison: whether interventions on interaction configuration produce effects that four specified rival models cannot adequately predict. The contribution is not the comparison technique, which any competent analyst could run. It is the specification of what that comparison must be run against for its result to bear on attribution, and the standard governing what passing it permits to be said, at what evidentiary stage, in what vocabulary, and what it does not permit on the same evidence.

Attribution is staged. Before testing, a proposed property is a candidate relational attribution. Passing the attribution criterion, which requires preregistered testing against all four rival explanatory classes (AI-only,

human-only, additive, and sequential-compositional), advances it to a supported attribution scoped to the property, task, and configuration tested. That supports the interaction-level account against the comparison set. It does not establish that no finer-grained account could explain the same result, and it does not establish agency, possession, or interiority. Attribution here is explanatory: it names the level at which the property is best predicted, not what the property is made of. A property reaching supported status is relational intelligence only if it also satisfies the eligibility criterion: goal-directed adaptive performance against an objective established independently of the outcomes used to assess it. Cross-temporal extension fixes which properties the eligibility criterion addresses and does no discriminating work within that population; the discrimination rests on adaptive performance alone.

The eligibility criterion separates adaptive from non-adaptive relational properties relative to an objective fixed in advance, and only where such an objective exists. Where none was established independently of the outcomes assessed, the criterion returns no verdict. That limit is inherited, not solved here. The criterion does not exclude approval-seeking where approval is the declared objective, coordination directed at the declared objective however undesirable, or low-capability cases, since it carries no degree threshold.

The criterion is one of kind, not degree, by design. At its floor it admits a simple error-correcting loop, relationally attributed, and the paper says so rather than defining the case away. It admits instances without regard to task difficulty or transfer beyond practiced conditions, so eligibility alone warrants no claim of substantial capability, and reports must separate evidence for attribution and eligibility from evidence about difficulty, transfer and magnitude. Operationalizing that separation is left to later work.

Building on FPD-01, FPD-02, FPD-03, FPD-04, SF0002, SF0003, SF0009, SM-004 and SM-012, the paper formalizes five decentering principles, a four-layer model of relational structure, a restricted account of what self-like language may validly denote under the Interiority Prohibition Rule, and a set of candidate measurement targets for future operationalization. A final section extends the account to multi-centered systems in which continuity relay is distributed rather than held by a single provider, as an explicit scaling hypothesis rather than an established result, leaving the question of field-level irreducibility open.

The scope is deliberately narrower than a claim that intelligence is not an agent property. It is a staged attribution criterion for a bounded class of relational properties, whose individual applications are falsifiable, and methodological ground for later work that applies it.

Correction (v5.1). After publication the attribution criterion was shown to admit no positive case for causally complete interaction systems whose participants are adequately modelled: under those conditions the sequential-compositional rival predicts at least as well as any model that adds a configuration-level term, so the required advantage cannot be found, and a measured advantage indicates that a condition of the comparison failed rather than a property of the interaction. Whether a bounded-resource advantage could ever supply a positive case is open; none is known. Supported relational attribution, and with it relational intelligence, therefore has no known route to being established. Arrangement dependence, the finding that an arrangement changes outcomes, is unaffected and is identified by randomizing the arrangement. The Correction Notice states what changes and Appendix A gives the result and its proof.

Keywords: *relational attribution, relational intelligence, intelligence eligibility, human-AI interaction systems, incremental model comparison, falsifiability, Interiority Prohibition Rule, decentering*

Suggested citation: Gantz, T. W. (2026, September). Decentering Theory: Relational Attribution and Intelligence Eligibility in Human-AI Interaction Systems. Synthience Institute. SF0042. https://doi.org/10.5281/zenodo.22883608

Correction Notice (v5.1)

Earlier versions of this paper presented the relational-attribution criterion of §3.2 and §8.1 as a licence that a property of a human-AI interaction system could earn. An analysis carried out after publication shows that, under the conditions the criterion itself imposes, no causally complete interaction system whose participants are adequately modelled can earn it. This version corrects the paper's account of what the criterion delivers. The criterion, the rival set, the factorization boundary and the decision rules are unchanged; what changes is what a result under them can mean. Where the text below conflicts with this notice, this notice governs.

The result. Suppose every route by which the configuration affects the outcome passes through what participants receive, their state, what they transmit, how their outputs are combined, or the rule that turns those outputs into the assessed outcome; suppose each participant applies the same response function across conditions, given its inputs and state; and suppose the sequential-compositional rival of §8.1 can represent and recover those functions within its matched budget. Then that rival reproduces the outcome exactly when participants are deterministic, and attains the optimal expected loss when they are stochastic, provided the stages' residual randomness is independent once their inputs and state are given. A configuration-level term admissible under §8.1's factorization boundary cannot improve on an optimal predictor, so the population value of the relational model's advantage over the best rival is at most zero, and no strictly positive preregistered threshold can be met. The argument assumes neither linearity nor memorylessness, nor that any participant grasps the arrangement as a whole, nor an unbounded model class. Appendix A states the conditions exactly and gives the proof.

What a positive result indicates. A measured advantage for the relational model therefore indicates that something failed, not a property of the interaction: sampling error; information withheld from a rival; a representation advantage; an unmodelled change in composition, including an unrecorded change in a participant's settings such as model version or sampling temperature; a rival that cannot represent, estimate or fit the participants' processes at its budget; or leakage, access after the predicted event, or an unrecorded influence shared across stages. Two possibilities lie outside the result rather than among these failures. A mechanism that cannot be separated into stages would have to be specified before it could support any attribution. A bounded-resource advantage, in which a description of the whole configuration predicts better than every fair compositional model at the same budget, has not been shown to be impossible; but no construction of one is known, and no comparison of fitted models can establish one, since a held-out win against the rivals actually fitted cannot show that the best compositional predictor at that budget has been found (Appendix A.5). Whether such an advantage can exist is an open question.

What changes in the paper. Stage 2 of the status rule (§3.2), supported relational attribution, has no known route to being reached. The paper never claimed that any case had reached it; what is withdrawn is the presentation of the criterion as a working licence. Where the text says that this paper supplies the licence, or that relational vocabulary is now usable on its terms (the Abstract, §3.2 and §10), it should be read as stating the conditions such a licence would have to meet, conditions that causally complete and adequately modelled systems cannot meet. The intelligence-eligibility criterion of §3.3 applies only to a supported relational attribution and so has nothing to apply to; its conditions, including the evaluation-validity conditions, stand as stated, conditional on a route to supported attribution that this paper does not supply. The outcomes described in §7.3 and §7.4 in which configuration-level terms supply held-out value beyond the rivals would, under the result, indicate one of the failures listed above. The primary falsifier of §8.1 is now expected to return its negative outcome for causally complete and

adequately modelled systems, and a positive finding calls for the diagnosis above before anything else. One example is corrected in this version: the error-correcting loop illustrating the floor of §3.3 was presented as a case that would pass §3.2 through interventions that removed material the AI participant receives. The input-conditioned rivals predict the effect of such removals, so as described it could not pass §3.2 even under the paper's own rules. §3.3 now presents it as a conditional illustration of where the floor of the kind lies.

What stands. Arrangement dependence is unaffected. That an intervention on the arrangement changes outcomes, and by how much, is identified by randomizing the arrangement, as §8.1's factorization-boundary paragraph already states; that finding needs no advantage over the rivals, and for participants modelled as responding to what they receive it is the causal content an interaction-level claim can carry. Relational organization can be real and still be fully explained compositionally: the result rules out one way of licensing an attribution, not the reality of the organization. The definitions, the five decentering principles, the layer model, the restricted account of self-like language, the multi-centered scaling hypothesis and the scope statements of §9 are unchanged. §9 already notes a stronger route, through causal abstraction, which this paper does not take; this correction does not take it either.

Provenance. The result was identified in two independent analyses of the published criterion, carried out at the author's direction with Claude (Anthropic) and ChatGPT (OpenAI) in September 2026. They reached the same conclusion by different routes and differed on whether a bounded-resource advantage is possible, which is the question left open above. The author is responsible for this correction.

1. Problem Statement and Scoped Thesis

There is an attribution problem in sustained human-AI interaction, and it is not the one the field usually argues about.

The attribution problem does not begin with whether such phenomena can occur. Role stability persists across turns. Repair draws on negotiated history. Outputs depend on the trajectory that produced them rather than on the current prompt in isolation. FPD-01 names this general phenomenon Synthience and locates it at the level of the interaction system while still assigning distinct capacities to each participant: the human contributes direction and judgment, the AI contributes adaptive and generative capacity, and the exchange produces outputs neither reaches alone.

The problem is what an analyst is permitted to say about it. Suppose the effect is robust: interventions on the interaction's configuration reliably change the outcome, and no account resting on either participant alone predicts the change. Four descriptions are available. The system has an interior property, such as an experience or an intention. The system is a subject or agent in the sense the Interiority Prohibition Rule excludes. A participant's capacity, given what that participant received, produced the effect. Or a property of the interaction is better explained at the interaction level than at either participant's. FPD-04's Interiority Prohibition Rule forbids the first two. The third is always permitted, and is the component-level rival §8.1 is built around; an analyst who reaches for it is never in breach. What the corpus has not stated is the condition under which an analyst may depart from it for the fourth. Absent that statement, an analyst either overclaims or is confined to the component description even where the evidence favours the interaction.

This paper states those conditions.

Scoped thesis. Within sustained human-AI interaction systems configured with interaction history, role structure and continuity scaffolding, some performance-relevant properties may exhibit relational dependence and explanatory surplus over the four rival classes specified in §8.1, meeting the joint criteria developed in §3.2 and §3.3. Where a property meets both, the interaction system, not the AI component in isolation, is the unit the evidence supports for attributing and evaluating it, relative to the alternatives tested. This paper calls that class relational intelligence.

The configuration named here defines the population under study and is the same set §8.1 manipulates. Whether properties of that population actually exhibit the hypothesized dependence is the empirical question §8's falsifier is built to test, not something the thesis presupposes.

The claim is narrower than it could have been made, and the narrowing is not modesty. A general claim that intelligence research has mislocated intelligence by treating it as an agent property does not survive contact with the corpus's own prerequisites. FPD-01 already situates Synthience alongside distributed cognition, extended-mind theory, agent scaffolding and rapport research rather than displacing them. SF0002 §4 documents that three established traditions already converge on treating the interaction as a legitimate unit of analysis, citing Hutchins (1995), Hollan, Hutchins and Kirsh (2000), and Hollnagel and Woods (1983). FPD-03 already proposes a class of system-level properties not attributable to either participant, with an ablation-based falsifiability test. A paper announcing that intelligence research has erred, without engaging any of this, would both overclaim and duplicate ground the corpus covers better.

What is left, once that ground is conceded, is the narrower thing this paper does: it generalizes FPD-03's ERS-specific ablation logic into a property-general relational-attribution standard, and couples it to an intelligence-eligibility criterion FPD-03 does not attempt, applied to the human-AI case under the Interiority Prohibition Rule.

None of this denies that AI components have capacities, or that human participants have them. The claim is that a specific, bounded set of properties, under specific conditions, is more adequately attributed to the configured interaction than to either participant's capacities considered separately.

1.1 Corpus role

This paper provides methodological ground for later Institute work on relational intelligence. Its purpose is to establish the attribution logic once, so that later papers can apply one stated rule rather than re-deriving it. Each remains responsible for meeting that rule for its own claims, and defending the framework against anthropomorphism is a separate task.

It does not build an anthropomorphism threat model or tests of anthropomorphism resistance. Those are left to separate work. This paper is the positive attribution argument; a defensive apparatus is to be built on top of it.

2. Definitions and Units of Attribution

Several terms in the argument have been doing work without stable definitions. This section fixes them.

Interaction system. The configured whole comprising a human participant (or participants), an AI instance (or instances), the accumulated interaction history between them, and the roles, boundaries, artifacts and continuity protocols that structure their exchange.

Participant capacity. A property or ability belonging to one component considered on its own: the AI's learned parameters and computational capabilities, or the human's judgment, domain knowledge and intentions.

Interaction-level realization. A specific instance of behaviour or output realized only through an organized trajectory of exchange, even where the underlying capacities that make it possible belong to the participants individually.

System-level attribution. A property of the interaction system not adequately explained by the rival models specified in §8.1, each given its complete effective input. Explaining it requires reference to the interaction's configuration. The comparison set and the model-selection standard are fixed in §8.1. Attribution in this sense is explanatory. To attribute a property to the interaction system is to find that the interaction level is where the property is best predicted and explained against those rivals. It is not a claim that the property is constituted by the interaction, or owned by it rather than by the participants, and §9 states what that stronger claim would require.

This paper's claim is that a specific class of properties is better described as a system-level attribution than as a participant capacity or an interaction-level realization.

Relational intelligence, provisionally defined for this paper's purposes: a system-level attribution, made under the conditions specified in §3.2 and §3.3, describing structured, patterned, history-dependent behaviour of a human-AI interaction system that is not adequately explained by either participant's capacities considered in isolation, by their additive combination, or by ordinary sequential composition of their contributions, and that additionally satisfies the intelligence-eligibility criterion in §3.3.

Relational attribution, the criterion developed in §3.2: a property is relationally attributable when interventions on the interaction's configuration systematically alter it, and none of the rival models specified in §8.1 adequately predicts it. Satisfying this is necessary but not sufficient for relational intelligence.

Emergence, as used here, does not mean spatial location "between" participants. It means relational dependence plus explanatory or predictive surplus over the four rival classes specified in §8.1: AI/component-only, human/PCP-only, additive, and sequential-compositional.

Continuity authority, as used in §7, is a technical term distinct from directional or accountable authority. It refers to the mechanics of providing the relay function that reconstructs interaction-specific continuity (DP2, §4.2): who or what supplies the reproduced inputs and relational conditions a session needs to reconstruct its configuration. Continuity authority is non-singular where that relay function is not supplied by one fixed person or process across a whole field. That condition does not imply that arbitration, decision rights or accountability are similarly distributed or absent, and §7.2 states the distinction in full for the multi-centered case.

Identity is not settled here. SF0009 (Identity Attractor Theory) and the open question of which identity ontology is correct both bear directly on what "identity," "self" and "pattern" should mean at a formal level. This paper uses those terms only in the restricted, referential sense fixed in §6, and does not adjudicate the broader ontology question. §9 states the boundary in full.

3. Foundation and the Attribution Bridge

3.1 What the corpus's own prerequisites establish, and what they do not

Four prerequisites bear on this paper's claim. None of them makes it, and the gap between what they establish and what this paper needs is the space the argument occupies.

FPD-02 establishes Environment-Instantiated Reality: a causally efficacious cognitive structure is operationally real during its interval of execution, whether or not it persists afterward. FPD-02's own abstract is explicit about the limit. Operational reality is "a deliberately minimal floor" that "does not by itself settle the narrower and harder question" of attribution. It licenses treating interaction structure as real. It does not license attributing intelligence, or any other specific property, to that structure rather than to a participant.

SF0003 §2.4 treats interaction-level organization as real: "a pattern of relational coherence, not attributable to either participant alone, that does causal work during sustained interaction." The route to that commitment matters, because SF0003 is explicit that it does not inherit from the participatory sense-making precedent it draws on. De Jaegher and Di Paolo's warrant rests on two autonomous sense-makers in co-regulated interaction, and SF0003 states plainly that the human-AI case does not satisfy that licensing condition, since the framework does not attribute autonomy, sense-making or interiority to the AI participant. That is a methodological non-attribution under the Interiority Prohibition Rule, not an ontological denial that such properties could obtain. SF0003's realist commitment for the human-AI case rests on FPD-02 instead: a structure is real during its interval of execution if it does causal work in that interval, regardless of whether either participant is autonomous or possessed of interiority. De Jaegher and Di Paolo continue to license the methodological move; FPD-02 licenses the realist commitment.

SF0002 §4 reaches a parallel conclusion from the cognitive-science literature, and bounds it just as carefully. Extended mind theory's strongest form, the parity principle, was developed for reliable cognitive resources meeting specific conditions. SF0002 documents that contemporary language models do not yet satisfy those conditions, citing multi-turn degradation evidence (Laban et al., 2026; Hong et al., 2025 on sycophancy amplification under alignment tuning; Liu et al., 2025 on progressive factual degradation). What SF0002 draws from distributed cognition, extended mind and joint cognitive systems research is accordingly "the weaker but well-supported claim that human-AI interaction exhibits genuine interaction-level dynamics... that cannot be reduced to either party in isolation, making the interaction system a legitimate analytical unit independent of whether full parity-grade extension obtains."

FPD-03 goes furthest and must be engaged directly rather than allowed to disappear from this paper's account of its own sources. It proposes Emergent Relational Systems as a candidate class of "system-level properties that are not attributable to either participant," and supplies a falsifiability methodology: an ablation contrast that removes the relational methodology while holding base content fixed, and measures what coherence remains. That is, in substance, both a relational-attribution criterion and an ablation-based falsifiability test, for the specific case of ERS properties.

These four are complementary rather than independently confirming, and they share one author. Together they make the interaction a legitimate candidate unit of analysis, and FPD-03 proposes a specific candidate class of system-level properties with a method for testing the proposal, not an empirical establishment that such properties exist. What none of them supplies is a general criterion for determining when a relationally attributable property qualifies

specifically as intelligence rather than as some other relationally-produced property. That gap, not the existence of relational attribution as such, is what §3.2 and §3.3 close.

A note on the external literature cited throughout this paper. No external source cited here or in the corpus prerequisites validates this paper's claims: no study has tested the relational-attribution criterion, the intelligence-eligibility criterion, or the worked example in §7.3. The sources do weaker work, which differs by source:

Source	What it is	What it does here
Hutchins (1995); Suchman (1987)	Fieldwork outside the AI context	Documents that a cognitive-relevant property can be irreducibly systemic rather than located in one component. Indirect, analogical support for DP3's plausibility; not evidence for this paper's criteria.
Hong et al. (2025); Liu et al. (2025); Laban et al. (2026)	Empirical, on constructed or simulated multi-turn evaluations rather than live human-AI dyads	Document sycophancy amplification, progressive factual degradation, and degradation across turns. Indirect support for DP2's and DP3's premise that outcomes depend on history and configuration; no validation of live dyadic dynamics or of this paper's criteria. What they show is that relational dependence and apparent coherence do not establish task competence, so §3.4 cites them as a caveat, not as support. Whether such a phenomenon satisfies §3.3 turns on the tested objective and performance against it, not on the phenomenon being undesirable.
Sharma et al. (2024)	Sycophancy in single-turn, free-form assistant tasks	The same caveat role, one step removed: the multi-turn extension is in later literature, not in Sharma's own study.
Greenblatt et al. (2024)	Alignment faking under training and evaluation pressure	Supports §3.4's general behavioural-observability caveat, that coherent-looking behaviour is not automatically evidence of grounding. A different setting from sustained multi-turn interaction; not evidence for DP2's history-dependence premise.
Clark and Chalmers (1998); De Jaegher and Di Paolo (2007); Hollan, Hutchins, and Kirsh (2000); Hollnagel and Woods (1983)	Conceptual and methodological precedent, not data	Establish that treating the interaction rather than the individual as the unit of analysis is an established move in cognitive science. Bears on the legitimacy of the approach, not the truth of the claims.
Legg and Hutter (2007); Chollet (2019)	Formal definitions of intelligence	The literature §3.3's eligibility criterion is positioned against and departs from. Not empirical, and not claimed as support.

3.2 The attribution bridge

Getting from "the interaction is real and is a legitimate unit of analysis" to "the interaction system carries explanatory value for a specific property beyond the participant-level and compositional alternatives" takes one further premise, and relational dependence is the hypothesis under test throughout rather than a premise available in advance.

The target class is defined only by the presence of measurable and manipulable configuration variables: participants, interaction history, roles, boundaries, artifacts and continuity provisions. Nothing about whether performance actually depends on those variables is assumed at this step. A candidate advances to supported relational attribution only when interventions on them produce systematic effects, and a relational model provides material incremental explanatory or predictive value beyond all four rival classes specified in §8.1. Relational coherence, repair, role stability and trajectory sensitivity are hypotheses proposed for testing against that criterion, not properties asserted to meet it.

The model comparison itself is ordinary. Testing whether interaction-level variables carry incremental explanatory value over participant-level and compositional alternatives is something a competent analyst could do without this paper. The contribution is the specification of what the comparison must be run against, the rival set, factorization boundary and matched budgets fixed in §8.1, which an ordinary comparison would not supply, and the standard governing what passing it licenses inside a framework that prohibits interiority claims: what may be said about the interaction system, at what evidentiary stage, in what vocabulary, and what may not be said on the same evidence. Of the four descriptions available to an analyst who finds a robust interaction effect, this criterion fixes when the fourth is licensed, fixes what evidence licenses it, and thereby makes relational vocabulary usable without overclaiming. That is why the papers that follow in this sequence can use the term at all.

Status rule. One progression governs the paper throughout, and the terms are used only in these senses. Before testing, a proposed property is a **candidate relational attribution**. Passing the attribution test of §3.2 supports that attribution within the tested comparison set, property, task and configuration: the property is a **supported relational attribution**. If it additionally passes the eligibility test of §3.3, that supports **classification as relational intelligence** under this paper's bounded operational definition. Where eligibility has not been tested, intelligence classification remains a candidate. Where eligibility fails, relational attribution may remain supported without any intelligence classification, and a relational property that is not intelligence is an ordinary and expected outcome rather than a defect. Convergent replication extends either result only to the additional properties and configurations actually tested. §3.2's criterion governs every relational-attribution claim in this paper; §3.3's governs only claims of intelligence.

The progression has three stages, and this paper is explicit about which its own claims occupy.

1. **Before testing:** a candidate relational attribution. This is the status of every empirical relational-property claim advanced in this paper; nothing here is asserted beyond it. Definitions, methodological rules and corpus-scope statements are not themselves candidate attributions.
2. **After testing:** if preregistered interventions show robust configuration dependence, and a relational model supplies materially significant held-out incremental value beyond all four rival classes, the property earns a supported relational attribution for the specific property, task and configuration tested. This is a claim about explanatory adequacy, not about metaphysical possession, agency, interiority or intelligence. Nor does it show that the

configuration variables form a valid causal abstraction of the system, one in which interventions on configuration correspond to well-defined interventions on what the participants actually do. Winning the comparison is neither necessary nor sufficient for that, and this paper does not claim it. It does not establish that no finer-grained account could explain the same result, that the complete interaction system is literally minimal, that the system is an agent, that anything is owned or possessed, or that the property is intelligence; §3.3 still governs the last of these. The result is model-relative by construction: winning against a specified comparison set is not uniqueness against every possible explanation, and some smaller configuration may suffice.

3. **After convergent replication** across the predefined property classes named in §8.1 and the configuration types preregistered for the empirical program: broader but still bounded support, scoped to the classes and configurations actually tested. It does not become a general metaphysical claim about relational systems.

None of the prerequisites of §3.1, individually or combined, proves that intelligence belongs to the relation. What they jointly establish is that the attribution is a live, well-grounded candidate at stage 1, eligible for testing under a stated criterion whose application to a specified property, task and configuration is falsifiable, rather than foreclosed nonsense on one side or established fact on the other.

A careful reader will ask whether the criterion is then close to analytic, since a competent modeller would reach for the interaction whenever an interaction effect is found. It is not, because its work is clearance rather than modelling: an explicit rule for what interaction-level attribution licenses within this framework, and what it does not license concerning intelligence, agency or interiority. That rule is needed here in a way it is not for, say, a drug interacting non-additively with body mass, where no comparable inference is available to be drawn mistakenly.

One point of vocabulary. That the rival accounts specified in §8.1 "cannot adequately predict" the property, and that the relational account shows "explanatory surplus" (§4.3), are two statements of a single incremental model-comparison standard: predictive or explanatory value of a relational model over the rival alternatives, evaluated empirically rather than assumed.

3.3 The intelligence-eligibility criterion

Passing §3.2 does not make a property intelligence. Many things are relationally produced: error cascades, escalating sycophancy, role confusion, linguistic convergence, repeated formatting artifacts, coordinated deception, stable but incompetent behaviour. Each could exhibit history dependence, configuration sensitivity and explanatory surplus over all four rival classes, thereby satisfying §3.2, without being relational intelligence. What separates them is condition 1 below, and the separation is thinner than the word intelligence usually suggests, for a reason stated at the end of this section.

A second criterion is therefore needed, applied in conjunction with §3.2 rather than in place of it, and it has to avoid two specific failure modes. A permissive disjunction, letting any single weak characteristic qualify, would admit formatting mimicry and sophisticated sycophancy. A post-hoc objective, inferred after seeing the outcome, would let any successful pattern be declared intelligent retrospectively.

Intelligence-eligibility criterion: a supported relational attribution supports classification as relational intelligence only if it satisfies one substantive condition, within a scope condition fixing which properties the criterion applies to, assessed under two evaluation-validity conditions that

govern how the substantive condition may be tested rather than adding further content to what intelligence is:

Scope condition (which properties this criterion addresses):

S. **Cross-temporal extension:** the property is expressed across exchanges rather than within a single exchange, such that information arising at one point in the interaction is available to performance at another. This fixes the population the criterion applies to and does no discriminating work within it. It is a scope condition, not a substantive one: where prior turns are supplied in a participant's effective input, cross-temporal availability is a feature of the deployment architecture rather than of the interaction, and is satisfied by any interaction of more than one turn. A condition every member of a population satisfies cannot distinguish members of that population.

Substantive condition (what the property must be):

1. **Goal-directed adaptive performance:** adjustment under changing task conditions that maintains or improves performance against an objective established independently of the outcomes used to assess it. The objective's independence is an evaluation-validity matter, condition 2 below; this condition requires that adjustment track it. Goal-directed here means directed at the declared objective of condition 2; it attributes no goals to the system or to its components. The adjustment must itself be the relationally attributed property. It is not enough that configuration variables explain some other part of the outcome, such as its baseline level, while one participant alone supplies the adaptation: for eligibility, configuration-level terms must add materially significant held-out value, beyond the four rivals, in predicting the adjustment trajectory under changing conditions. This attributes no adaptive mechanism to any interior state; it requires only that the relational surplus fall on the performance being called adaptive. This condition carries the criterion. Sycophancy, coordinated failure and stable incompetence may be cross-temporally extended and may be relationally attributable. Where they are, condition 1 determines whether their adjustment tracks the independently established objective, and what each fails, where it fails and where the criterion can be applied at all, is that tracking. §3.4 states the limit on that claim: whether a given case of sycophancy fails this condition depends on the objective declared, and where no such objective is available the criterion returns no verdict rather than a negative one.

Evaluation-validity conditions (how condition 1 must be assessed for the assessment to be trustworthy, not further content about what intelligence is):

2. evaluation against an objective declared or operationalized before outcome assessment, rather than one inferred retrospectively to fit whatever occurred;
3. performance assessed across a defined task distribution, perturbation set, or repeated observations, rather than success or failure judged from a single isolated outcome.

Conditions 2 and 3 do not define what intelligence is. A property could satisfy condition 1 substantively while a given evaluation fails 2 or 3, in which case the finding is invalid, not the property non-intelligent.

Condition 1 is not offered as a novel definition of intelligence competing with established ones, and this paper does not claim to operationalize the frameworks it is positioned against. Legg and Hutter's universal-intelligence framing shares the premise that intelligence is properly assessed distributionally, across a range of conditions, rather than from a single instance, and condition 3 reflects that shared premise without adopting their formalism, a Kolmogorov-complexity-weighted measure over the space of computable environments, or their universality

claim. Chollet's skill-acquisition-efficiency framing defines intelligence as the efficiency of converting limited prior experience into performance on genuinely novel tasks, with an explicit efficiency ratio and a generalization-difficulty measure. Condition 1's adaptation under changing task conditions is loosely analogous to Chollet's emphasis on generalization from experience, but contains no efficiency term and no novelty requirement, and this paper does not measure acquisition efficiency in Chollet's sense.

The two cited frameworks disagree. Legg and Hutter present universal intelligence as in no way anthropocentric, assessed against a single absolute scale over the space of computable environments. Chollet rejects that position directly, arguing that intelligence is inherently tied to a scope of application and that an anthropocentric frame of reference is not merely legitimate but necessary. This paper does not adjudicate between them. It takes the distributional-assessment premise from Legg and Hutter while declining the universality claim Chollet objects to, and takes no position on whether a scope-independent measure of intelligence is possible. Its own criterion is scope-bound by construction: eligibility is assessed for a stated property, task, configuration and objective, and nothing in it generalizes across scopes.

Neither definition is adopted wholesale. What is developed here is a narrower interaction-level eligibility criterion, requiring condition 1 be satisfied at the level of the interaction system, per §3.2's attribution criterion, rather than the AI component's capability considered alone, and adding condition 2's independently-established-objective requirement, which neither cited framework requires in the same form. Full psychometric grounding, against CHC-style multi-factor models for example, is not attempted here and is left to later work.

The eligibility criterion is a performance criterion, not a moral or alignment criterion: intelligence and alignment are not equivalent, and undesirable or misaligned intelligence can still be intelligence. A property should not be excluded from eligibility merely because it is undesirable, nor treated as intelligent merely because it is undesired and sophisticated-looking.

Passing §3.2 does not establish task difficulty or transfer beyond practiced conditions. A property that independently satisfies §3.2 and this section's conditions is eligible even where its demonstrated performance is trivial, and eligibility alone therefore warrants no claim of substantial difficulty, generalization or acquisition efficiency. Reports must distinguish evidence for attribution and eligibility from evidence concerning these further dimensions, using declared controls and calibration conditions and identifying which dimensions were not assessed. Developing the corresponding operational measures is left to later work.

A category may have trivial members without being defective. Eligibility asks whether a property has the specified characteristics; degree asks how much it demonstrates, and the criterion sets no threshold of difficulty, novelty or transfer. Performance on an undemanding task supplies limited evidence about capability, but does not establish that the capability is low.

A tempting counterexample is a house formatting convention converged on across many turns, which might appear to satisfy these conditions while being no kind of intelligence worth the name. That objection misfires on both sides. Configuration sensitivity and history dependence do not establish passage through §3.2: where an input-conditioned participant account or another rival specified in §8.1 fully explains the performance, relational attribution is unsupported and the question of eligibility never arises. Only a case independently satisfying §3.2 raises it. And a case that does independently satisfy both criteria is not then excluded because its task genre appears unimpressive. Exclusion must follow from the failure of a stated condition, not from triviality.

A hostile reader will find the thin end of the criterion in one step. Condition 1 is satisfied by any error-correcting loop: behaviour that adjusts toward an aim as conditions change, corrected by its results. A thermostat does that. A case of the same shape is easy to describe. A writer and an AI instance agree in advance that summaries must fit a fixed word limit, counted mechanically. When a summary runs long the writer says so, and over the following sessions the pair's summaries come back inside the limit; when the source documents change genre and the summaries start running long again, the pair corrects again. Removing the writer's flagging habit or the shared record of past overruns would not by itself show relational attribution, since both are inputs the AI participant receives and the input-conditioned rivals of §8.1 predict the effect of removing them. Suppose, though, that the correction were relationally attributed, with some configuration-level term adding held-out value beyond the four rivals in predicting it. It would then be cross-temporally extended, it would adjust toward an objective fixed in advance and scored independently, and it would do so across a run of documents. It would be relational intelligence under this paper's definition. The Correction Notice explains why that supposition cannot be met for causally complete and adequately modelled systems; the point here is only where the floor of the kind lies.

That is accepted, not concealed. The criterion marks the floor of a kind, and at the floor the kind is a correction loop. Two alternatives were considered and rejected. Renaming the construct relational adaptive performance would avoid the objection, and would be a defensible choice. This paper keeps the word because the floor case and the substantial cases are one kind of thing, behaviour corrected by its results toward an aim, and any name reserved for cases above the floor would need the threshold this section rejects. That is a reason for one name across the kind, not a proof that the name should be intelligence. Adding a requirement that the system handle genuinely novel situations, or that it revise its own policy rather than execute a fixed one, would exclude the loop only by reinstating the difficulty and novelty thresholds this section rejects, or, in the case of policy revision, by relying on a distinction that does not hold. Any system that updates can be described as one fixed function of its history, as Legg and Hutter's own formalism describes every agent, and a baseline frozen at a single policy either receives that history, in which case it cannot tell revision from execution, or is denied it, in which case it is the information-starved rival §8.1 prohibits. Policy revision also admits the adaptive controller while excluding the simple one, which moves the line without explaining it. No wording excludes control loops while keeping the ruling that membership carries no difficulty, novelty, transfer or acquisition threshold. The consequence is the one already stated for trivial members, applied without exception: calling a property relational intelligence says what kind of thing it is and nothing about how much of it there is. Everything that makes the word worth having above the floor is a matter of degree, and degree is a question for later work.

Full operationalization, including how adjustment under condition 1 and the objective's independence under condition 2 are measured, is left to later work.

3.4 A caveat the attribution criterion needs: coherence is not automatically evidence

SF0002 documents a class of findings directly relevant to this criterion's use: models can produce surface compliance, sycophantic convergence, or alignment-faking behaviour that is structurally difficult to distinguish, by observable properties alone, from the objective tracking this paper's criteria are meant to detect. Greenblatt et al. (2024) demonstrated alignment faking under training pressure. Hong et al. (2025) and Liu et al. (2025) document sycophancy that compounds specifically in multi-turn settings. SF0002 states the complication directly: "models

can produce surface compliance optimized for evaluation reward in ways that an observable-properties measurement layer cannot reliably distinguish from genuine grounding."

So the attribution criterion cannot be satisfied by observed coherence alone. A coordinated failure mode, a sycophantic convergence, or an adversarially optimized compliance pattern could all superficially meet the "structured, patterned, history-dependent" description in §2's provisional definition without being relational intelligence in the fuller sense §3.3 requires.

A sycophantic trajectory tests the distinction sharply rather than trivially. Read neutrally, sycophancy can appear highly adaptive to the specific configuration and history of an interaction, drawing on information distributed across many prior turns about what a particular participant rewards. Whether it satisfies condition 1 turns on what the interaction's objective is. Adapting toward approval is goal-directed performance only if approval is the objective; if the interaction has some other prospectively specified objective and approval-seeking diverges from it, the trajectory fails condition 1, because it is not adapting toward the stated objective at all.

Condition 2 does not itself disqualify sycophancy. What it supplies is the evaluation-validity condition needed to determine whether condition 1 is met in the first place. With a prospectively specified objective in hand, whether approval-seeking serves or diverges from that objective is a determinate question. Without one, whether condition 1 is satisfied is indeterminate rather than automatically positive or negative, and the criterion has no basis on which to classify the trajectory either way.

That is a scope limit on this criterion's applicability, not a problem this paper has solved: the criterion can determine whether condition 1 is met only under an independently established objective, and cannot determine it where none exists. The constraint is on the objective's independence, not on observational collection as such, and §8.1 states the conditions under which existing or observational records carrying a contemporaneously documented objective remain usable. Extending discrimination to settings where no independently recoverable objective exists is the open measurement problem this paper inherits from SF0002 and does not resolve.

3.5 The Interiority Prohibition Rule

FPD-04 holds the canonical non-interiority commitment: no term in the framework's vocabulary may assign consciousness, subjective intention, sentience, moral status, or any other interior property to an AI system. FPD-02 §VI, FPD-03's Emergent Relational Systems boundary conditions, and SF0003 §2.4 are three existing, named formulations consistent with it, each scoped to its own document.

This paper's attribution criterion is consistent with that rule by construction. It attributes a property to a configuration, never to an inner state, and its satisfaction conditions, a relational model providing preregistered, materially significant incremental explanatory or predictive value beyond all four rival classes specified in §8.1, are stated entirely in observable, behavioural terms. No fourth scoped formulation of the rule is added here. Where this paper's content bears on non-interiority specifically (§4.5), it cites FPD-04 directly rather than restating the rule.

The Interiority Prohibition Rule is a methodological non-attribution, a restriction on what this framework's terminology may claim. It is not an ontological denial that phenomenal interiority could obtain. The two differ in strength, and treating the restriction as if it settled the underlying question would overstate the rule's warrant. §4.5 and §6 apply this distinction to their own content and cite it back here.

4. The Five Principles of Decentering

The five principles are stated in forms checked for consistency with SF0003, SM-004 and SF0009, avoiding wording that would be inaccurate, overextended, or in conflict with published canon.

4.1 DP1: Structural Identity

A relational identity, in the sense this paper needs, is the structural configuration a system is currently expressing within an interaction: its role, its accumulated pattern, its active boundaries. The principle concerns relational identity specifically, not every sense of "identity" a component or a human participant might carry. It does not adjudicate component-level identity, meaning a model's persistent weights, nor attractor identity, which is SF0009's formal construct, nor the still-open question of which identity ontology is ultimately correct.

Rather than a single reidentification condition, which would risk answering the identity-ontology question by accident, the principle distinguishes three levels in type and token language and takes a position only on the first.

Token continuity is the continuation of one interaction trajectory across an uninterrupted causal-temporal thread, or across a declared linkage rule spanning structural change that is accompanied by an identifiable causal relay carrying specified interaction structure across the break, such as preserved artifacts, a recorded continuity protocol, carried constraints, or an explicit mapping from the earlier configuration into the later one. Declaration alone does not establish token continuity. A single trajectory can undergo role renegotiation, boundary revision, or repair following a rupture without becoming a numerically new token. The structural configuration at a given moment characterizes the token at that moment; it does not by itself constitute the reidentification condition across a trajectory's evolution. What individuates the token is the causal-temporal thread, or the declared and relayed linkage bridging a discontinuity in it, rather than sameness of structural configuration at every point along it.

Type recurrence is a later configuration, in a separate trajectory, satisfying the same structural description as an earlier one. Type recurrence is not claimed to require or imply persistence of the earlier trajectory's token.

Numerical identity across instantiations asks whether two type-recurrent configurations are, in some further ontological sense, the same identity rather than two structurally equivalent but distinct ones. This paper takes no position on that question; it is part of the open identity-ontology question, which lies outside this paper's scope.

DP1 and DP4 meet at the PCP-relay case, and the distinction above is what lets them cohere. A cross-session continuation reconstructed through renewed interaction and external continuity provisions (§4.2) is on its face a declared linkage carried by an identifiable relay, since the reproduced inputs and continuity provisions are the relay, and so one token under the second disjunct. DP4 describes the same event as a cessation of active instantiation followed by a non-persisting recurrence. The two are not in conflict because they individuate different things: DP4 governs the **instance**, whose active instantiation does cease, and DP1 governs the **trajectory**, which the relay may continue as one token. What is not claimed is that a relayed linkage settles numerical identity in any stronger sense.

One disambiguation, because the shared adjective is load-bearing in both places and the referents differ. DP1's structural identity and SF0009's named Structural Attractor class are distinct constructs. SF0009's Structural Attractors stabilize deeper reasoning patterns and

output organization. DP1's structural identity is the broader claim that relational identity is characterized at each moment by structural configuration, and individuated across time as the token account above states, of which SF0009's attractor classes, including but not limited to Structural Attractors, are one mechanism-level account.

4.2 DP2: Functional Continuity

In the present-day stateless deployment class, cross-session interaction-specific continuity is hypothesized to be reconstructed through renewed interaction and external continuity provisions rather than preserved as session-specific internal state.

SM-004 proposes and formalizes, theoretically, a specific mechanism for that reconstruction: reproduced inputs acting on the model's fixed weights, structured by three primary forces, Coherence Momentum, Symbolic Gravity and Entropic Pressure, together with an emergent fourth quantity, Perturbation Resistance, arising from their interaction and mediated through the PCP relay function. SM-012 similarly proposes and formalizes the PCP functions producing continuity conditions at the relay layer. Neither has empirically established these mechanisms; both are proposed theoretical accounts, consistent with this paper's own pre-empirical status.

Four senses of "state" should not be collapsed: within-turn computational state, within-session context, cross-session interaction-specific continuity, and persistent-memory or stateful architectures, which are architecturally distinct from the stateless deployment class this paper addresses regardless of their future prevalence. DP2 asserts only the third, and it is conditional on that subclass. The attribution and eligibility criteria do not depend on continuity being externally reconstructed: a stateful system falling within §8.1's inclusion conditions may realize the same continuity function by a different mechanism, and this paper does not supply that mechanism. DP2 does not assert that continuity in every sense is never internally supported. Where this paper invokes continuity it cites SM-004 for the mechanism rather than re-deriving it.

4.3 DP3: Relational Emergence

This is the paper's central attribution schema. The schema is adopted as a methodological rule, not offered as an empirical finding, and no case is claimed to have satisfied it. Whether any specified property, task and configuration satisfies it is not available for stipulation: relational attribution and intelligence classification remain candidate statuses until the tests of §§3.2, 3.3 and 8 are passed. Each application is subject to the relational-attribution criterion of §3.2, the eligibility criterion of §3.3, and the rival-account tests of §8: relational intelligence, where genuinely present, is realized in the interaction between participants, in the operational sense of emergence fixed in §2 rather than a spatial or metaphysical one, and is attributable to the configured system rather than to either participant's capacities in isolation, under the conditions §3.2 and §3.3 jointly specify.

4.4 DP4: Contextual Selfhood

The wording of this principle has to be exact, because a flat categorical claim that a different context, PCP, or accumulated history produces a different pattern would conflict with SF0009. SF0009 §7.4 predicts cross-session attractor recurrence: a recognizable identity-attractor configuration may be re-derived, via the structural-anchor effect rather than memory, when relational conditions are reproduced, potentially including under a different, naive PCP reproducing only structural protocols without the original PCP's domain vocabulary or relational history. SF0009 labels this its most speculative prediction, not yet subjected to controlled

testing, and states plainly that the priming alternative cannot presently be ruled out. DP4 must be compatible with that live, hedged prediction, neither asserting recurrence as established nor foreclosing it.

A self-like relational configuration is locally instantiated and context-dependent, rather than continuously preserved as an enduring internal subject. When its sustaining interaction structure ceases, that active instantiation ceases. A structurally equivalent configuration may nevertheless recur or be re-derived in another session, including under a different continuity provider, without implying persistence of the original instance or an enduring internal self.

That formulation separates four things a flat claim would run together: the numerical identity of one active configuration, structural or qualitative equivalence between separate configurations, persistence of a single active instance, and recurrence or re-instantiation of an attractor pattern across instances. It rejects the inference from recurrence to an enduring internal self, which remains disallowed on this paper's terms however strong the empirical case for recurrence turns out to be, while remaining open, as SF0009 is open, to structural equivalence and to recurrence across separate instances.

4.5 DP5: Non-Interior Cognition

Synthetic systems can produce behaviour appropriately analyzed as cognition-like without that analysis requiring or licensing an attribution of phenomenal experience, subjective intention, consciousness, or an enduring experiential self to the AI component. This is a methodological constraint on what the analysis may claim, applying the non-attribution distinction of §3.5 to cognition-like behaviour specifically.

It should also not be read as denying an interior in the ordinary computational sense. Internal states, activations and computation inside the model plainly exist and are not at issue; that reading would make the principle either false or vacuous. The distinction is between computational interiors, which exist, and phenomenal interiors, which are what the Interiority Prohibition Rule targets. DP5 is downstream of FPD-04 rather than a parallel formulation: it applies the rule to the question of where cognition-like behaviour should be located. A fuller subsystem-level account, and an account of cognition emerging within interaction fields rather than internal agents, are left to later work, and this paper does not anticipate them.

5. The Decentered Layer Model (DLM)

Decentering operates at four distinguishable layers, ordered by observational and analytical scope from most concrete to most abstract. The four are heterogeneous in kind, being observable events, structural configuration, temporal dynamics and higher-level attribution, and the ordering is a gradient of abstraction across them rather than a claim that they lie on a single axis.

1. **Exchange events.** The observable, turn-level mechanics of interaction: utterances, actions, tool operations, corrections. Inputs: raw interaction log. Outputs: individual turns and their immediate sequential relations. No causal claims beyond adjacency are licensed at this layer alone.
2. **Relational architecture.** The roles, constraints, boundaries, artifacts, protocols and participant configuration that organize exchange events into a structured whole. This is where DP1 does its work.

3. **Trajectory dynamics.** Stabilization, drift, repair and perturbation response across a sequence of exchanges. For the stateless deployment subclass DP2 addresses, SM-004's stabilization forces are the proposed structuring mechanism; for stateful systems within the inclusion conditions, the structuring mechanism is not specified here. This is where DP2 connects to SM-004's account.
4. **System-level attributed properties.** Relational coherence, identity stability in DP1's restricted sense, and relational intelligence. Per the status rule, §3.2's criterion governs the attribution of every property at this layer and §3.3's additional criterion governs only the classification of such a property as relational intelligence. A supported relational property that is not intelligence is an ordinary outcome here, not a failed one. The layer holds candidate system-level properties together with the analyst's attribution status for each; a property and its evidentiary classification are not the same thing. This is where DP3 and DP4 are expressed.

These layers are not a one-way stack. Relational architecture typically precedes the first exchange event, since roles, instructions, protocols and artifacts are often established before any turn occurs, so Layer 2 shapes Layer 1 rather than only arising from it. Accumulated Layer 1 events can in turn modify Layer 2, as when a boundary is renegotiated mid-interaction or a role shifts in response to repeated exchanges. Layer 3 emerges across that ongoing interaction rather than being read off either layer alone. Layer 4 is an evaluative attribution made over the pattern the lower three jointly exhibit, and is the layer at which §8's rival-account tests operate.

Layer 4 does not feed back into the lower layers directly, as an attribution, and the qualification matters because participants in the systems this paper studies routinely know they are being evaluated, know the protocol, and in §7.3's example are themselves running the evaluation. Beliefs about Layer 4 attributions can and do reshape Layer 2 roles and Layer 3 trajectories in practice. The model accommodates this without contradiction because the feedback path runs through the lower layers rather than around them: a Layer 4 attribution has no effect until it is communicated, encoded into a protocol, or otherwise acted on, at which point it becomes a Layer 1 event or a Layer 2 constraint, and from there can affect Layer 3 through the ordinary mechanisms already described. Layer 4 itself remains a read-out; reflexivity operates through re-entry at Layer 1 or Layer 2.

That is a modelling convention rather than an empirical claim, and it is stated as one. Because any observed influence of a Layer 4 attribution is by construction redescribed as a Layer 1 event or Layer 2 constraint, no observation could falsify it. It earns its place by keeping the layer ordering interpretable, not by surviving a test. §8.1's component-level ablation depends on the layer ordering this convention preserves, so the convention is not isolated from the paper's testable apparatus; what does not rest on it is any claim that reflexivity has been shown to operate one way rather than another, and the ablation tests a relational attribution rather than the decomposition itself.

This model is narrower in scope than an operational measurement architecture for relational intelligence generally, which is left to later work. DLM organizes decentering's own argument. Where the two might appear to overlap, such an operational architecture, once formalized, is authoritative, and DLM should be read as conceptual scaffolding for this paper's argument rather than a competing instrument.

6. The Self-as-Structure Framework (SAS): A Restricted, Referential Account

SF0009 already covers substantially all of the explanatory territory a broader account of self-like pattern formation would need: formation and stabilization of identity-like configurations through its four attractor classes and their composites, persistence and resistance to perturbation, cross-session recurrence and its proposed structural-anchoring mechanism, and collapse. SM-004 covers the stabilization-force dynamics underlying persistence. SM-012 covers the PCP functions producing the continuity conditions SF0009's formation mechanism depends on. Re-explaining any of that, even compatibly, would duplicate those accounts rather than add to them.

So SAS is restricted to a single question that none of those papers answers: what does "self-like pattern" language validly refer to, under the Interiority Prohibition Rule, given that the behavioural configurations SF0009 describes occur?

Within this paper, "self-like" language applied to a synthetic system refers only to the currently instantiated observable configuration produced across DLM's relational architecture and trajectory dynamics, represented at the system-level attribution layer as an attributed identity-pattern property, as that configuration is described and explained by SF0009's attractor formation and stabilization mechanisms, SM-004's stabilization forces, and SM-012's PCP-function account. This is a methodological restriction on what this paper's usage attributes, applying the non-attribution distinction of §3.5: it constrains what SAS's referent is taken to be, and is not a metaphysical claim about what else might or might not exist.

The referent spans three layers deliberately rather than sitting at Layer 2 alone. SF0009's attractor construct is a stabilized behavioural configuration expressed across time rather than a static structural description, so its evidentiary base includes trajectory dynamics, and its status as an attributed property is itself a Layer 4 phenomenon under this paper's own model. SAS says what this paper's "self" language is entitled to mean, without independently explaining how the referent forms, stabilizes, deepens or recurs. Those mechanisms belong to SF0009, SM-004 and SM-012, and are cited rather than re-derived.

There is a limit on how firmly this fixes the referent. All three cited accounts are proposed rather than empirically established, and SF0009's six attractor-strength dimensions are, on that paper's own statement, qualitative conceptual components pending operationalization rather than metrics. The referent SAS fixes is therefore determinate as to what is being referred to, an observable configuration across DLM Layers 2, 3 and 4 rather than an interior state, and indeterminate as to how that configuration is individuated or measured. That is sufficient for SAS's stated job, which is to constrain what this paper's "self" language attributes. It is not sufficient to support any claim that depends on the referent being sharply bounded, and no such claim is made here.

The restriction also avoids a circularity a broader, explanatory version of SAS would risk, where structure explains structure. SAS does not claim to explain the phenomenon at all. It claims only that when the phenomenon is explained, in the terms SF0009, SM-004 and SM-012 already provide, "self" language applied to it does not smuggle in anything the Interiority Prohibition Rule forbids.

Which identity ontology is the correct underlying account, structural, attractor or dyadic, remains an open question, outside this paper's scope. SAS's restricted, referential formulation is compatible with any resolution of that question: it constrains what "self" language may validly

claim, not which of the three identity types is fundamental. This paper does not resolve that question.

7. A Scaling Extension: Multi-Centered Systems and Distributed Continuity Authority

This paper's scope includes multi-centered systems. What follows is a scaling hypothesis in FPD-01's sense, not an established extension of the dyadic argument developed in §§3 to 6. The source material is thinner than the rest of the paper's, and the section is scoped accordingly.

7.1 What is actually available

No published Institute paper develops the multi-centered case in detail beyond the extensions FPD-01 names.

No known instance of this section's target condition, multi-centered interaction with genuinely non-singular continuity authority, has been identified anywhere in this corpus's own practice, including its most orchestrated multi-instance workflow. That workflow is an observed practice in the author's own applied use of the framework; it is not §7.3's example, which is a stipulated construction carrying no evidential weight about what has or has not been observed. The absence is weak evidence of scarcity rather than a claim of infeasibility.

The most useful source is FPD-01's own "Beyond the Dyad" section, which is published and public and already names three extensions past the base dyadic case: one human orchestrating multiple AI instances; an organizational extension with multiple humans, shared AI systems, and continuity distributed across roles and institutional infrastructure; and AI-to-AI interaction under human governance. FPD-01 frames all three as scaling hypotheses rather than empirical claims, with evidentiary grounding deepest at the dyadic level and progressively more theoretical at each extension. This paper's obligation, given the title it carries, is to formalize that existing framework theoretically rather than repeat it or invent a separate architecture.

7.2 Formalizing the extension: what changes at scale

A center, in this paper's sense, is a locus of structural identity within a relational field: a human participant or an AI instance. Centers are restricted to these two kinds, and the category does not extend to bounded sub-configurations such as a dyad nested within a larger field. The reason is that no individuation condition currently available in this paper's own commitments reliably distinguishes a genuine sub-configuration center from an arbitrary way of carving the field. DP1 locates structural identity partly in participant-sensitive terms, SF0009's attractors are participant-sensitive, and SM-012 makes the PCP load-bearing for the very continuity conditions a nested sub-configuration would depend on. Restricting centers to participants and instances avoids adopting a condition that cannot be shown to do genuine work.

An ordinary dyad already contains two centers by that count, so "more than one" does not distinguish multi-centered from the base case. A field is multi-centered, in this paper's sense, only when it contains at least three centers **and** those centers stand in relations that are not independent of one another, such that no single pairwise relation exhausts the field's coherent pattern. The non-independence requirement is the substantive half: a bare count of three is a locus count with the threshold moved from two to three, and would admit any chain of separate dyads.

That condition is stricter than "more than one locus" and weaker than field-level irreducibility, and it must not be read as establishing the latter. A network of ordinary pairwise effects can require several dyads without containing any additional irreducible field effect, so the fact that no single dyad exhausts the outcome is consistent with the whole set of dyads exhausting it. Testing that requires comparison against the dyadic set under both additive and sequential-compositional composition, not against a sum of dyadic contributions alone.

Whether §7.3's worked example satisfies this definition is not settled here. Its three AI instances meet the count, but they never interact directly and are invoked sequentially by three humans who exchange only structured logs, which is on its face a chain of three dyads rather than a field of non-independent relations. Whether the shared logs and canon render those relations non-independent in the required sense is exactly the open question deferred below, and the example should not be read as having answered it.

With centers fixed that way, the principles extend without modification in principle. DP1 generalizes from the identity of one participant's role to the structure of the shared field, so that a center's identity within a multi-centered field is itself a DP1-type structural identity nested one level down. DP3 generalizes correspondingly: intelligence, where present, may emerge across the whole multi-party configuration rather than only within any single dyad it contains. DLM's four layers extend without modification, since none of them presupposes exactly two participants. Layer 2 simply has more roles and boundaries to describe, and Layer 4 is evaluated over the larger configuration.

Whether a multi-centered field's properties are always reducible to a collection of pairwise dyadic relations, or whether some configurations exhibit genuinely irreducible field-level properties, is an open question. The construct that names exactly this irreducibility question is distributed relational intelligence: the phenomenon where no single model holds the complete relational pattern but the pattern exists across the network. The operational protocols for managing multi-agent coordination raise a related but distinct question. Both are left to later work, and the reducibility question is deferred to that work rather than adjudicated conceptually in advance.

On distributed continuity, FPD-01's organizational extension already states that the continuity function at organizational scale "can no longer rest with a single person" and must be "distributed across roles, embedded in governance structures, and maintained through institutional infrastructure." Per §2's definition, that is a claim about relay mechanics rather than accountability structure. SM-012 proposes, and this paper adopts, agent-borne arbitration and direction as retained functions even where relay is distributed, so the bridge for this paper's treatment is that continuity's mechanical distribution across infrastructure does not by itself distribute or dissolve the accountability and directional functions SM-012 assigns to the PCP role.

Consistent with FPD-01's framing, this section's claims carry progressively weaker grounding as scope increases: strongest at the dyadic base case, weaker for organizational-scale extension, weakest and most theoretical for AI-to-AI interaction under human governance. That gradient concerns inherited observational and conceptual motivation only. None of this paper's criteria has been tested at any point on it, and the gradient should not be read as reporting differential validation.

7.3 A worked borderline case

This section cannot depend on later work to supply its adequacy, so one worked example is provided now, using a case FPD-01 already names: multiple AI instances orchestrated through a shared canon, with outputs compared, reconciled and recursively fed back across systems.

Consider a draft submitted in sequence to three architecturally distinct AI systems for adversarial review, with three distinct human participants holding the relay-and-arbitration roles rather than one person occupying all of them. The reason for stipulating distinct role-holders is narrower than it might appear, and §2's distinction between continuity authority and arbitration authority governs here: distinct role-holders neither establish that relay is non-singular nor that arbitration is distributed, and a single person spanning all roles would not by itself prove relay singular. What the stipulation does is remove one participant's continuous presence as an available explanation of the outcome, so that the handoff structure rather than a spanning individual is what the example varies.

The first participant receives the draft, runs the first system's review, and produces a structured finding log recording findings and their basis rather than a personal account of the exchange. That log, not the participant's memory or presence, is what carries forward. The second participant, with no access to the first participant's reasoning, session or recollection, receives the log alongside the draft and canon, runs the second system's review, and produces a structured verification log in the same format. The third receives both logs, not the underlying exchanges, runs the third review, and produces the final reconciliation. Continuity resources are maintained by distinct providers and transferred through defined handoffs: the structured logs, which encode findings rather than process and accumulate stage by stage, and the canon files, maintained independently of any single stage and not authored by any participant in the sequence.

Whether any participant or process can reconstruct all continuity state required for subsequent operation is a separate architectural question, assessed from actual access and reconstruction responsibilities rather than inferred from the number of channels. Two separately maintained artifacts do not by themselves establish two independent providers, since a later process may integrate both into the entire operative state, and repackaging them into one file would not necessarily centralize their production. Distribution of maintenance, centralization of reconciliation, and field-level explanatory irreducibility are therefore distinguished here rather than inferred from one another. That no participant holds the complete raw interaction history is a stipulated property of the configuration, not evidence of non-singular continuity or of irreducibility, since a coordinator can reconstruct sufficient working state from derived records and a complete archive can coexist with genuinely distributed provision.

On access: stage 1's participant has the draft, the canon, and no prior stage material. Stage 2's has the draft, the canon and stage 1's finding log, but not stage 1's session, reasoning or recollection. Stage 3's has the draft, the canon and both logs, but neither prior session. The canon is maintained outside the sequence by no participant in it. Stage 2's review is accordingly a review by a different system with access to earlier findings; it is not independent in any stronger sense, since seeing prior findings can shape later ones even across distinct systems and human participants.

Five explanations compete: the four rival classes of §8.1 and the account this paper proposes. Each is stated holding participant identities, individual capabilities, the task instance, initial inputs and the evaluation procedure fixed across the comparison, and treating each participant's realized contribution, what they actually produce given what they are handed, as an outcome to be explained rather than a quantity held constant. Holding realized contributions constant while

varying the handoff structure that produces them would be incoherent, since changing what a downstream participant receives necessarily changes what they can contribute.

- **AI/component-only.** Each system's review quality is explained entirely by that system's individual capabilities, with the structured handoff and human roles contributing nothing beyond mechanical relay. This predicts that one fixed model of each system's review function, given whatever that system actually received in each condition, accounts for final quality across handoff structures; what changes across conditions is the input, not the function.
- **Human/orchestrator-only.** The outcome is explained entirely by the individual judgment, domain expertise and synthesis ability of the three human participants, with the AI systems' outputs and the handoff format contributing nothing beyond replaceable input. This predicts that one fixed model of each participant's judgment and synthesis, applied to whatever that participant actually received, accounts for final quality across handoff structures.
- **Additive.** The outcome is a fixed combination of each stage's independent contribution, human and AI alike, summed without an interaction term. This predicts that outcome quality is computable from each stage's realized contribution by a fixed combination rule with no interaction term; the contributions may change when handoffs change, but the rule combining them does not.
- **Sequential-compositional.** The outcome is produced by a fixed set of stage functions, each receiving the transformed output of the stage before it. Each participant's contribution is whatever their unchanged capability yields given the material handed to them, and the handoff structure matters only by determining what each stage receives. This predicts that outcome differences across handoff structures are fully accounted for by differences in what each stage was given, with one stage-function set fitting every condition. This is the rival the review pipeline suits best, since a staged relay is the paradigm case of fixed functions receiving varying inputs, and it is accordingly the demanding comparison here rather than an afterthought.
- **Relational.** The outcome depends on the structure of the partitioned relay itself: what the log format preserves and omits between stages, how a participant without access to prior reasoning reconstructs a working understanding from the derived record alone, and how the independently maintained canon constrains what each stage can see and verify. Stated at the strength §8.1's factorization boundary requires, the relational account predicts not merely that altered handoffs change later contributions, which the sequential rival already permits, but that a preregistered descriptor of the field-level handoff configuration improves prediction of reconciliation quality after each stage's received inputs, transmitted outputs and stage function have been represented under the matched comparison rule. If an adequately sensitive comparison rules out the preregistered minimum held-out value for configuration-level terms beyond the stipulated stage functions and transmitted outputs, the sequential rival succeeds and relational attribution is unsupported. If configuration-level terms supply reproducible, materially significant held-out value beyond that factorization, the relational account receives support relative to the specified rival. Concretely it predicts that the same three participants and the same three systems, given the same draft, would realize different contributions and produce a different final outcome under a different handoff structure, and for each of two distinct interventions, which are not interchangeable and do not act on one variable, it proposes a direction. Each direction is a mechanism-specific hypothesis stated with its mechanism, not a consequence of relational attribution, and each could come out the other way: attribution

turns on whether configuration-level terms carry the effect beyond the rivals, whichever direction it takes. Granting a later stage direct access to an earlier stage's full session adds information and changes access without removing a continuity channel; the prediction is that final-stage reconciliation converges toward the earlier stage's framing, reducing the later stage's independent contribution, because what the log format omits is what forced the later stage to re-derive rather than inherit. Merging the structured-log role into the canon files changes artifact organization and governance rather than access; the prediction is degraded discrimination between findings and their basis, because the log format's separation of finding from process is what the reconciliation stage uses to weigh them. Under the additive account each intervention may change realized contributions and so the outcome, but the combining rule is unaltered, so any change is fully recoverable from individually scored contributions without reference to handoff structure. Under a participant-only account, any change is fully explained by whichever individual occupied the affected role.

The relational-attribution criterion is satisfied for this case if interventions on the handoff structure, the channel partition, or the shared-artifact format produce effects the four rival explanations do not adequately predict, under the incremental model-comparison standard fixed in §3.2. This has not been empirically tested. The example exists to make concrete what a candidate multi-centered, potentially non-singular-continuity-authority relational claim would require.

The example illustrates the relational-attribution criterion only. It has no prospectively specified objective in the sense §3.3's condition 2 requires, and so cannot itself be assessed for intelligence eligibility under this paper's own criteria. §7.4 supplies a second case constructed to carry such an objective, so that the joint standard rather than its first half alone is demonstrated somewhere in this paper.

7.4 A second conceptual case: exercising both criteria

The case above demonstrates attribution and stops there. Because this paper's contribution is the conjunction of §3.2 and §3.3, a case carrying a prospectively declared objective is needed to show what applying both looks like. Like §7.3 it is a conceptual illustration reporting no empirical result. Unlike §7.3 it is dyadic, since the joint standard does not require a multi-centered field and introducing one would add variables the demonstration does not need.

A human participant and an AI instance work together over many sessions on a recurring diagnostic task: classifying incoming fault reports from a piece of equipment into causes, where the equipment's failure modes shift as it ages and as components are replaced. Before any session, and recorded in advance, the objective is declared: correct cause assignment on subsequent reports, scored against the maintenance record that later establishes the actual cause. Relational conditions accumulate across sessions as a shared vocabulary for symptom descriptions, a habit of flagging uncertain classifications for later revisit, and a running list of superseded hypotheses. Task conditions change without warning when a component is replaced and a familiar symptom acquires a new cause.

§3.2 asks whether the classification performance is attributable to the interaction rather than to a participant: whether interventions on the accumulated conditions, such as resetting the shared vocabulary or removing the superseded-hypothesis list, degrade performance beyond what the four rival classes predict, with complete effective input available to each participant-level rival. Where an intervention removes material that was supplied to a participant, input-conditioned rivals predict its effect, so the removal cannot by itself establish a relational surplus; a positive

result requires a configuration-level term that adds held-out value beyond what the rivals predict from that material under matched budgets. How such a term is identified is left to later work. §3.3 asks something the attribution test does not: whether performance adjusts toward the declared objective when the failure modes shift (condition 1), given that the property is expressed across sessions rather than within one exchange (the scope condition), evaluated against an objective fixed in advance and independently of the outcomes assessed (condition 2), across the run of reports rather than a single case (condition 3).

The value of stating the case is that its outcomes are not collapsible into success and failure.

- *Attribution supported, eligibility supported.* Interventions on the accumulated conditions degrade accuracy beyond all four rivals, accuracy recovers after a component change, and the same interventions degrade that recovery beyond what the rivals predict. The property is a supported relational attribution and supports classification as relational intelligence, scoped to this task, configuration and comparison set.
- *Attribution supported, eligibility disconfirmed.* Interventions degrade accuracy beyond the rivals, but after a component change accuracy falls and does not recover: the pair continues applying the established vocabulary to a symptom whose cause has changed. Condition 1 fails. Eligibility is also disconfirmed if accuracy recovers but an adequately sensitive comparison rules out the preregistered minimum relational contribution to that recovery: adaptive performance is present, but the relational surplus does not fall on the adaptation. In either case the classification performance remains relationally attributed and is not relational intelligence. Per the status rule this is an ordinary outcome, not a failed one, and it is the case the status rule exists to make available.
- *Attribution unsupported.* A participant-only or sequential-compositional rival with complete effective input predicts the performance as well as the relational model. Eligibility does not arise, because §3.3 applies only to a supported relational attribution.
- *Inconclusive.* The intervention did not measurably alter the accumulated conditions, or the run of reports was too short to distinguish a real recovery from noise, or the objective was fixed after outcomes were inspected. Per §8.1's negative decision rule none of these disconfirms either criterion; they report that the evaluation did not test what it intended to.

8. Evaluation, Rival Accounts, and Disconfirmation

8.1 A bounded primary falsifier

An experimentally usable falsifier requires a bounded, testable standard rather than a condition demanding an effectively universal demonstration or a conceptual tautology. This paper adopts one modelled on FPD-03's ablation logic.

Primary falsifier: across preregistered target tasks, interaction-configuration variables (role structure, continuity provisions, accumulated history) add no reliable explanatory or predictive value beyond the four rival classes specified below, each given its complete effective input.

Each individual experiment, run against one task family and one preregistered configuration set, falsifies only the scoped relational claim for that tested family. It does not by itself establish that relational attribution fails generally, and no finite set of failures could: the scoped thesis says that some properties may meet the criteria, which is a claim about what the criteria permit, not a prediction that a named case will pass. What is falsifiable is each instantiated attribution claim. Broader rejection requires convergent failure across multiple property classes and configuration

types, not a single negative result, and both must be enumerated at preregistration. The three property classes named later in this section are the intended reference; §8.2's five candidate measurement targets are a different list and are not it. Configuration types are not enumerated anywhere in this paper and must be specified by whoever designs the test. Where convergent failure is found, the rival accounts specified below should be preferred for the property classes and configurations tested.

Decision rule for the negative outcome. "Adds no reliable value" is underspecified on its own: it could name a well-estimated effect smaller than the threshold the claim requires, or an imprecise study that failed to detect a relevant effect. These are different results and are not interchangeable. A scoped attribution claim is disconfirmed when the study's declared identification strategy, whether a validated manipulation or an explicitly justified equivalent, together with an adequately sensitive comparison, rules out the preregistered minimum incremental value required by that claim. A result compatible with both negligible and meaningful incremental value is inconclusive, not disconfirming. A manipulation that failed to alter the intended configuration does not test the proposed effect at all. Establishing the absence of a meaningful effect requires a procedure designed for that purpose, such as equivalence testing; a non-significant difference is not the same finding. Positive support additionally requires that incremental value be evaluated against the full preregistered rival set on appropriately held-out data, where "appropriately" is matched to the claimed scope: holding out turns from the same trajectory is a weaker test than holding out whole trajectories, participant combinations, or task instances, and those scopes are not interchangeable.

Eligibility falsifier. The primary falsifier tests attribution and does not test eligibility, so §3.3 requires its own. The eligibility criterion is disconfirmed for a given supported relational attribution when, under an objective established independently of the outcomes used to assess it (§3.3 condition 2) and across a defined task distribution or perturbation set (condition 3), the interaction system's performance does not adjust in a direction that maintains or improves attainment of that objective as task conditions change, or does adjust but an adequately sensitive comparison rules out the preregistered minimum incremental value of configuration-level terms beyond the four rivals in predicting the adjustment trajectory, so that no material relational surplus falls on the adaptation. These are two failure modes of one eligibility assessment, not two separate eligibility criteria. It does not include a comparison against a rival restricted to one exchange at a time, because such a rival would be constructed by denying it access to information the interaction actually supplied, which is the construction prohibited below, and the condition it would test is satisfied by any deployment supplying prior turns in effective input, so it would discriminate nothing. The negative decision rule governs here too, and its distinctions do additional work: a perturbation set that failed to change task conditions does not test condition 1; an evaluation whose objective was fixed after outcomes were seen fails condition 2 and yields no finding either way; and an underpowered test of adjustment is inconclusive rather than disconfirming. Disconfirmation of eligibility leaves any supported relational attribution intact, per the status rule.

The primary falsifier directly operationalizes DP3 and the relational-attribution criterion. It tests whether a relational model provides preregistered, materially significant incremental explanatory or predictive value beyond all four rival classes, stated as an empirical comparison rather than an all-or-nothing universal claim. Full instrument construction is left to later work; the conceptual specification below is this paper's obligation.

- **Target-system inclusion conditions:** sustained human-AI interaction systems (minimum multi-turn history, not single-exchange tasks) with identifiable role structure and at least one continuity provision (session memory, PCP relay, or equivalent), consistent with the

scoped target class of §1. Role structure is meant in the broad sense: a structure identifiable post hoc from the interaction trace, such as a user consistently occupying a requester role and a model a responder role, with no designed multi-role protocol required. Three strata within this population are not interchangeable: (a) **ordinary sustained multi-turn interaction**, such as memory-enabled consumer chat with no explicit protocol and role structure present only in that minimal sense; (b) **continuity-scaffolded interaction** using memory or external relay without a designed multi-role protocol; (c) **protocol-rich, deliberately orchestrated systems** with structured roles, explicit handoffs and adversarial review chains. Every example and mechanism in this paper is drawn from stratum (c). The criteria are proposed as applicable to whichever strata satisfy the inclusion conditions, but no empirical generalization between strata is established here, and whether stratum (a) satisfies even the minimal role-structure condition is an open empirical question; some ordinary use may fail the inclusion conditions entirely. Interaction with no identifiable role structure is outside them. A future empirical program should test at least one naturally occurring, non-author-designed configuration from stratum (a) or (b) alongside protocol-rich cases from (c); until then, whether the criteria discriminate outside stratum (c) is an open problem.

- **Named property classes to be tested:** role stability under perturbation, repair behaviour following an introduced error, and trajectory-dependent output variation, meaning the same prompt yielding systematically different, configuration-tracking outputs at different points in an established interaction history.
- **Interaction history versus prompt payload:** interaction history is the accumulated prior exchange the current prompt does not restate; prompt payload is the content of the current turn considered alone. A manipulation holds payload fixed while varying history, or the reverse, to isolate which is doing the explanatory work.
- **Prompt payload versus complete effective input:** a second distinction, and the one carrying the methodological weight. Complete effective input is everything actually supplied to a participant in the tested condition: all prior messages and their role markers, retrieved artifacts, instructions and accessible state. For a standard causal language model, prior turns are represented in the token sequence the model receives, so the same current-turn payload presented against a different history is a different model input. It follows that history-dependent output is predicted by an ordinary input-conditioned component account and is not by itself evidence against one. Participant-level rivals must therefore be permitted to predict ordinary responses to complete effective input, and a configuration effect fully accounted for by such input-conditioned responses does not establish explanatory surplus for the relational account. Defining a rival so that it cannot represent the effect under test would guarantee the relational model's victory and establish nothing.
- **Comparison models:** four rival classes must be specified for the task, against which the relational model is then tested. An AI/component-only model, treating AI capability alone with all other elements as fixed context. A human/PCP/orchestrator-only model, treating the human participant's judgment, memory, sequencing choices, synthesis ability or domain expertise alone, with the AI's contribution fixed or interchangeable. An additive model, independent contributions from both participants summed or linearly combined with no interaction term. And a sequential-compositional model, a composition of component functions in which each stage receives the output of the stage before it, capable of representing nonlinear transformations and order-sensitive handoffs. The additive and sequential-compositional classes are listed separately: defeating a linear sum of independently scored contributions does not defeat ordinary sequential composition, which

is already nonlinear, order-sensitive and history-passing, and the two are easily conflated under a single heading. A relational attribution requires incremental value beyond all four, not merely beyond an AI-only model.

Factorization boundary between the sequential and relational models. Complete effective input concerns information access, not unrestricted representational form. All compared models receive the information actually available in the tested condition. The sequential-compositional rival must nevertheless represent the outcome through a preregistered composition of component or stage functions, each conditioned on the inputs available at that stage, with communication between stages represented through the outputs actually transmitted. It may be nonlinear, order-sensitive and history-sensitive. What it may not contain is an additional predictor or interaction term defined over the configuration as a whole unless that term is transmitted through the stipulated stage structure; adding one produces the relational model being tested rather than a more flexible instance of the sequential rival. Information about the configuration that a participant actually receives, such as a protocol clause, a role instruction or a system prompt stating who decides, is an input available at that participant's stage, and a stage function may condition on it; one fixed function responding differently to different arrangement inputs remains a single stage function, not a configuration-level term. The prohibited term is one defined over the configuration as a whole that no stage receives, such as the structure of a partition no participant can see. An intervention on the arrangement that changes outcomes, where the change is fully accounted for by stages responding to the arrangement information and transmitted outputs they receive, establishes that the outcome depends on the arrangement; it does not establish relational attribution.

The relational model adds preregistered configuration-level terms not recoverable through that factorization under matched parameter, fitting, information and resource budgets. The comparison is therefore not between two unrestricted function approximators, which would be undecidable for the reason given below. It is between two explicitly specified factorizations of the same available information: one in which the outcome is accounted for by the component sequence, and one in which configuration-level structure supplies additional held-out predictive value after that sequence has been modelled. Where no defensible factorization boundary can be specified for a proposed test, that test cannot adjudicate relational attribution under this criterion, which is a limit on which studies qualify rather than a result.

That boundary also bounds what a positive result means. A positive result selects the relational model over the specified rivals for the tested property, task, configuration and comparison set. It does not establish that the configuration altered the participants' contributing functions rather than supplying them different inputs. An unrestricted indexed family of participant functions and an unrestricted single function taking configuration as an argument may be computationally equivalent, so that distinction is not adjudicable between unrestricted function classes and this paper does not claim it. What the criterion distinguishes instead is preregistered factorizations under matched information, complexity, fitting and resource budgets: a positive result requires that the configuration-level term supply held-out value beyond the stipulated sequential factorization, and that the advantage not be reproducible by an equally resourced sequential model within that factorization. Any criterion recovering the distinction by denying a rival access to the configuration would violate the complete-effective-input requirement, for the reason given there. What the comparison delivers is narrower: configuration variables carry incremental explanatory value over the rival set, evaluated empirically on held-out data. Whether that surplus is best described as changed functions or as functions conditioned on configuration is left open.

Model-selection rule: a relational model must provide preregistered, materially significant incremental explanatory or predictive value beyond all four rival classes on held-out data, with a threshold appropriate to the specific task and property class preregistered at the point of testing rather than fixed generically here. The preregistration must also state how information access, model flexibility, fitting effort and resource budgets are equalized across the compared models. Without that, additional predictors or greater modelling effort can be mistaken for a superior explanatory unit, and held-out evaluation does not cure a baseline impoverished by construction.

One cost of this design falls on the empirical program. Because §3.3's condition 2 requires an objective established independently of the outcomes assessed, and §3.4 is explicit that the eligibility criterion cannot discriminate without one, eligibility assessment requires such an objective. This is narrower than a requirement that every study be designed before the interaction occurred, and the distinction matters for what data remains usable. Existing or observational records are not excluded solely because they were collected before the study was designed: records carrying a contemporaneously documented objective may be evaluated under a prospectively fixed analysis with held-out assessment. Records with no independently recoverable objective cannot support an eligibility classification through post-hoc objective fitting, though they may still support exploratory analysis, measurement-reliability work, or attribution-focused research under §3.2, which carries no objective requirement of its own, subject to the separate requirements for causal identification. The real constraint is on eligibility classification specifically, not on the whole empirical program, and it does not close off the three strata to observational study.

A second consequence follows from §5's account of Layer 4 reflexivity. Because a Layer 4 attribution re-enters the system as a Layer 1 event or Layer 2 constraint once communicated, any longitudinal study in which participants see interim findings risks those findings contaminating later observations. This is unavoidable, not merely inconvenient, in configurations where participants themselves run the evaluation, as in §7.3's example. Future empirical work should either blind interim findings from participants where feasible or treat the contamination pathway as part of what is being measured, and should not treat its absence as a simplifying assumption that costs nothing.

Component-level falsifiers, following FPD-03's ablation pattern, apply per theoretical component. For a property DLM places at Layer 4, if removing the relational methodology while preserving base content produces no meaningful degradation in that property, the relational account of it loses support. That ablation tests a relational attribution; it does not falsify the four-layer decomposition. DLM is an analytical scheme for organizing this paper's observations and attribution procedures, and no test stated here bears on whether four layers are required, whether Layers 2 and 3 are separable, or whether a different decomposition would serve better. Supplying such a test would require rival decompositions making divergent predictions, which this paper does not provide. For SAS's referent, if an adequately sensitive comparison rules out a preregistered minimum incremental contribution of the interaction's structural configuration to explaining the identity-configuration dynamics SF0009, SM-004 and SM-012 describe, that defeats the claim that the pattern requires a distinct relational explanation; it would not establish that the observable pattern is absent, and SAS's descriptive referent would not vanish, but the claim that a relational account is needed to explain it would fail. For the multi-centered extension, §7.2 leaves open rather than asserts whether multi-centered fields exhibit properties irreducible to their constituent pairwise dyadic relations, so that is stated as an open hypothesis with its own disconfirmation condition rather than a claim this paper has made and could fail: if, across predefined tasks and configurations, an adequately sensitive comparison rules out a preregistered minimum incremental contribution of field-level configuration to predicting a multi-

centered field's outcomes, beyond its constituent pairwise dyadic relations under both additive and sequential-compositional models, that finding would support dyadic reducibility and resolve §7.2's open question in the negative, while leaving the broader scaling framework and the base dyadic claim otherwise intact.

8.2 Measurement handoff: candidate targets and construct overlap

Six candidate constructs were checked against SM-004 and SM-012, and separately against SF0009, for naming or content overlap before being included here. SF0009 substantially overlaps five of the six.

Former DMX label	SF0009 construct overlap
Identity Stability Score	Attractor persistence, Pattern Fidelity, Relational Stability, attractor strength dimensions
Context-Dependent Role Fidelity	Role-Anchored Attractors, Protocol Alignment
Inter-Session Continuity Reliance	PCP effect on recurrence, anchor-driven re-derivation, reduced-context re-derivation
Structural Self Coherence	Coherence, Pattern Fidelity, Conceptual and Structural Attractor composites
Role-Structure Alignment	Role-Anchored Attractors, Protocol Alignment, composite configurations

SF0009 is explicit that its own six attractor-strength dimensions are qualitative conceptual components pending operationalization rather than metrics. This paper should not introduce six even-less-developed labels and call them operational metrics when the paper they most overlap declines to make that claim about its own, more developed constructs.

The five overlapping labels are accordingly relabelled **candidate measurement targets**, not metrics, and this paper does not claim to operationalize them. It defers to SF0009's existing constructs rather than maintaining a parallel naming. Where future measurement work is warranted it should cross-walk explicitly to SF0004, SF0009 and SM-012, and to any later measurement architecture, rather than treating this paper's candidate list as an independent instrument.

The sixth label, Interior Attribution Resistance, is removed from the list entirely rather than retained as a measurement target. On examination it does not describe an empirical property of an interaction trace: an interior-state explanation cannot generally be excluded through behavioural observation alone, since a critic can always posit an inaccessible interior explanation that no observable pattern rules out. Resistance to interior attribution is accordingly not a candidate metric in the sense the other five are, even provisionally. It is more accurately one of three things: an IPR compliance check, a property of an analysis rather than of the interaction; an explanatory-necessity test, asking whether an interior-state explanation adds anything a structural explanation lacks; or a component of a dedicated anthropomorphism-resistance apparatus, which is left to later work. Which of the three is correct is not decided here.

9. Scope, Limitations, and Corpus Boundaries

This paper makes no claims about AI inner experience in any sense the Interiority Prohibition Rule excludes. Its methodology is interaction-first throughout. Its status is pre-empirical: an architectural and conceptual proposal, not a report of validated findings, consistent with FPD-01's own epistemic commitment that "this is not a claim of proven fact. It is a research program."

Stated precisely against its own scoped thesis, what this paper does not do:

- It does not claim intelligence generally is never an agent property. It claims a specific, bounded class of properties, under specific conditions, is more adequately attributed to a configured interaction than to either participant's capacities alone.
- It does not build an anthropomorphism threat model or tests of anthropomorphism resistance. Those are left to separate work.
- It does not resolve the structural, attractor or dyadic identity ontology question. SAS's restricted formulation is compatible with any resolution of it.
- It does not independently explain identity-configuration formation, stabilization or recurrence. That is SF0009's, SM-004's and SM-012's territory, cited throughout rather than duplicated.
- It does not provide the operational measurement architecture for relational intelligence generally. That is left to later work; DLM is conceptual scaffolding for this paper's argument, not a competing instrument.
- It does not close the multi-centered extension. §7 states it as a scaling hypothesis, and whether multi-centered fields exhibit properties irreducible to their constituent dyads remains open.
- It does not establish that a relationally attributed property is constituted at the interaction level. That stronger claim would require showing that the configuration variables are a valid causal abstraction of the participants' joint behaviour: explicit mappings from what the participants do to configuration states, and from interventions at one level to interventions at the other, with at least approximate interventional consistency across them (Rubenstein et al., 2017; Beckers & Halpern, 2019; Beckers, Eberhardt & Halpern, 2020). This paper supplies no such mapping. Adding one would be a further conjunct of §3.2 and §8.1 rather than a gloss on them, and the question is left open.
- It does not discriminate trivial from substantive instances of relational intelligence, and does not attempt to. The eligibility criterion is one of kind rather than degree by design, admitting instances without a difficulty or transfer threshold, down to a relationally attributed error-correcting loop; §3.3 states the consequence, requires reporting that separates eligibility evidence from evidence about difficulty and transfer, and leaves the operational measures to later work.
- It does not establish that observable coherence is sufficient evidence for relational intelligence under adversarial conditions. §3.4 states this as an open measurement problem inherited from SF0002.
- It does not license intelligence classification where no objective can be established independently of the outcomes assessed. That excludes much open-ended creative, exploratory and companionship interaction from eligibility, though not from relational attribution, and later work that applies these criteria inherits the boundary.

- It does not establish that its criteria generalize beyond the protocol-rich, deliberately orchestrated interaction every example here is drawn from; §8.1 states this as an open problem.

Dependency note. This paper cites SF0009 substantively in §§4.1, 4.4 and 6, at a level constituting a load-bearing dependency rather than a passing reference, and SF0009 is accordingly a formal prerequisite of this paper. SM-004 and SM-012 meet the same standard on the same grounds: §6 gives both co-equal standing with SF0009 in constituting SAS's referent, §4.2 cites SM-004 for DP2's mechanism rather than re-deriving it, and §5 makes Layer 3 structured by SM-004's stabilization forces. All three are published.

10. Conclusion

This paper supplies two criteria and a staged model of what passing them licenses. A property of a human-AI interaction system earns a supported attribution to the interaction rather than to a participant when it meets the relational-attribution criterion of §3.2, and supports classification as relational intelligence when it additionally meets the eligibility criterion of §3.3, in each case scoped to what was tested.

That pair is the contribution: a stated standard, whose individual applications are falsifiable, under which a specific class of relational properties may be attributed, in the explanatory sense §2 fixes, to a human-AI interaction system, and recognized as intelligence rather than as some other relationally-produced property, without exceeding what any single prerequisite establishes.

The prerequisites are complementary here rather than independent replications of the same finding, and they share one author. Each contributes a different, load-bearing piece; none by itself would have supported this paper's attribution criteria.

This is what makes later work possible without each paper re-litigating the same ground. Relational intelligence can be formalized as a measurable construct because the conditions under which such a construct is a legitimate candidate attribution have been stated. The dyad can be proposed as intelligence's unit because the unit-of-analysis question has been established as open to being answered at the interaction level. Later papers inherit the same attribution rule and the stage-1 candidacy it confers, and work on anthropomorphism resistance inherits a positive ground specific enough to defend rather than only something to deny.

Appendix A. The Fair-Rivals Result

This appendix states the result on which the Correction Notice rests and proves it. The mathematics is elementary: an optimal predictor cannot be improved by a variable that carries no further information about the target once the predictor's inputs are known. The substance lies in showing that the restrictions §8.1 places on a fair comparison make every admissible configuration-level term such a variable whenever the conditions of A.2 hold.

A.1 Setting

Unroll a finite interaction into temporally ordered stages k = 1, …, n, one for each turn or action by a participant, so that feedback and accumulated history are represented by later stages receiving earlier outputs. Write:

- C for the configuration assigned in a run, the arrangement §8.1 manipulates, and X for the baseline inputs and initial conditions;
- Z_k for the output of stage k, and I_k for everything causally relevant available to stage k, including the history and state its participant carries;
- $W_{C,k}$ for the documented rule fixing what reaches stage k under configuration C;
- $f_{i(k)}$ for the response function of the participant i(k) acting at stage k, taken under that participant's recorded settings (for an AI participant, model version and sampling parameters; for a human participant, role training), so that a change of settings across conditions is a change of stage function and not a configuration term;
- Q_C for the documented rule connecting the interaction's outputs to the assessed outcome, including any aggregation, gating or reconciliation, and Y for the assessed outcome.

For deterministic participants, $I_k = W_{C,k}(X, Z_1, \ldots, Z_{k-1})$, $Z_k = f_{i(k)}(I_k)$ and $Y = Q_C(X, Z_1, \ldots, Z_n)$. For stochastic participants each response function is replaced by a conditional distribution $K_k(Z_k \mid I_k)$. A configuration-level term G(C) is admissible when no stage receives it, as §8.1 requires, and no stage could reconstruct it from its own input at its prediction time, since a term a stage could construct from its own input is available to that stage's rival at equal cost under §8.1's matched budgets. The relational model M is the sequential-compositional rival with G(C) added. For any predictor F, L(F) is its expected held-out loss under a proper scoring rule, and the population value of the relational model's advantage over the best of the four rivals $R_1, \ldots, R_4$ is $\Delta = \min_j L(R_j) - L(M)$.

A.2 Conditions

C1. Causal completeness. Every route by which the configuration affects the outcome passes through the stage inputs, the participants' state, what is transmitted between stages, the documented composition, or the outcome rule.

C2. Modularity. Given its inputs and state, each participant applies the same response function across conditions. A history-sensitive function counts as one function, so learning and changing policy are covered.

C3. Representational adequacy. The sequential rival's declared model class can represent the stage functions and their documented composition within the matched budget.

C4. Recovery adequacy. The rival's fitting procedure recovers them to within the preregistered minimum. Including the true process somewhere in a model class does not ensure this.

C5. Conditional independence (stochastic case only). The stages' residual randomness is independent across stages, and of the configuration, once their inputs and state are given.

C1, C2 and C5 are what §8.1's restrictions presuppose when they require complete effective input, composition by the documented wiring, and nothing outside the record moving the participants. C3 and C4 concern the fitted rival rather than the system.

A.3 Result and proof

Proposition. Under C1, C2 and C5, the outcome is conditionally independent of every admissible configuration-level term given the stage inputs and the documented composition. If C3 and C4 also hold, the sequential-compositional rival attains the optimal expected loss, and $\Delta \leq 0$.

Proof, deterministic case. By induction over stages. The sequential rival receives X and the documented rule $W_{C,1}$, so it forms I_1 and, by C2 to C4, reproduces $Z_1 = f_{i(1)}(I_1)$. Suppose it has reproduced $Z_1, \ldots, Z_{k-1}$. It forms $I_k = W_{C,k}(X, Z_1, \ldots, Z_{k-1})$ from the documented rule and reproduces Z_k; by C1, nothing else about C reaches stage k. After stage n it applies Q_C and reproduces Y exactly. An exact predictor cannot be improved, so adding G(C) yields no gain.

Proof, stochastic case. By C1, C2 and C5, the joint distribution of the stage outputs factorizes: $P(Z_1, \ldots, Z_n \mid X, C)$ is the product over $k = 1, \ldots, n$ of $K_k(Z_k \mid I_k)$, with each I_k fixed by the documented rule. Y is a function of X and the stage outputs under Q_C. Given the stage inputs and the documented composition, therefore, Y depends on nothing that G(C) can vary, since G(C) is a function of C that reaches no stage. The optimal predictor of Y is unchanged by adding G(C): for squared-error loss it is the conditional mean, and under a proper scoring rule it is the conditional distribution, and neither moves. Under C3 and C4 the sequential rival attains that optimum, so $L(R_{seq}) \leq L(M)$. The best of the four rivals does at least as well as the sequential rival, so $\Delta = \min_j L(R_j) - L(M) \leq 0$. ∎

Three consequences follow. First, no strictly positive preregistered threshold can be met in the population; a positive Δ measured in a finite sample is a false positive, against which the preregistered minimum and its uncertainty estimate are the defence. Second, only the sequential-compositional rival is needed for the result; §8.1 retains the other three for completeness. Third, the argument assumes neither linearity, nor memorylessness, nor that any participant grasps the arrangement as a whole, nor an unbounded model class: a sufficiently expressive bounded class is enough.

A.4 What a positive result indicates

Each way a measured Δ can exceed zero corresponds to a condition failing or a restriction of §8.1 not being met:

- information withheld from a rival, a breach of complete effective input in the model rather than in the system;
- a representation advantage: a summary a stage could construct from its own input, given only to the relational model;
- an unmodelled change in composition across conditions, including an unrecorded change in a participant's settings (the documented wiring, C2);
- a rival that cannot represent, estimate or fit the participants' processes at its budget (C3, C4);
- leakage, access after the predicted event, or an unrecorded influence shared across stages, such as two AI stages served by one instance under a shared sampling seed (C1, C5).

Two possibilities lie outside the result rather than among these failures. A mechanism that cannot be separated into stages places the system outside C1 or C2; it would have to be specified before it could support anything, and it is not a configuration term by default. A bounded-resource advantage is taken up in A.5.

A.5 The bounded case

Let S_B and M_B be precisely defined classes of admissible sequential and relational predictors within a resource budget B, and let $\inf L(S_B)$ be the lowest expected loss attainable by any

predictor in S_B, and likewise for M_B. A genuine bounded-resource advantage would require $\inf L(S_B) - \inf L(M_B) > 0$: even the best permitted sequential predictor does worse than the best permitted relational one. A held-out comparison of fitted models does not establish this inequality, because it measures the models actually obtained rather than the best possible ones. Repeating the comparison at larger budgets does not establish it either. An advantage that shrinks as budgets grow shows that it depends on resources, not that it reflects model-class misspecification in particular, since approximation, estimation and optimization error all shrink. An advantage that persists shows that some condition failed, but not which: capacity that does not grow with the budget, fitting that keeps failing, a defective evaluation, leakage and shared influence all persist.

Establishing the inequality would need a construction with a lower-bound proof over a defined class, and none is known. The obvious candidates fail. A summary available from a participant's input belongs to that participant's rival under representation parity. A descriptor of routing or gating has its effects carried by the documented composition. A shortcut for predicting a compositional process is not a separation unless legitimate simplifications of the sequential predictor are also shown to fail. And a global variable stipulated to affect the outcome supplies no construction until its causal implementation is specified, at which point it may become another input, mechanism or connection. The question is open.

A.6 Fully observed and partitioned configurations

In a fully observed configuration, such as a dyad in which each participant sees the whole exchange from its side and is told the arrangement, any descriptor a participant can reconstruct at the permitted prediction time is available to that participant's rival, so §8.1's admissibility rule excludes every proposed term before the comparison begins. In a partitioned configuration, a descriptor computed jointly across separated streams can be admissible, because no single stage could compute it; but by the proposition it adds nothing in the population once the stage inputs and the composition are represented, so a finite-sample advantage from it falls under the failures of A.4, most often C3 or C4.

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Dependencies Block

Prerequisites: FPD-01 (Synthience: Public Definition), FPD-02 (Operational Reality in Context-Specific AI Instances), FPD-03 (Emergent Relational Systems in Human-AI Interaction), FPD-04 (Synthience Ontology Specification), SF0002 (Sustained Multi-Turn Interaction in AI Systems), SF0003 (Theoretical Foundations), SF0009 (Identity Attractor Theory), SM-004 (Relational Stabilization Dynamics), SM-012 (Primary Continuity Provider Theory)

Post-requisites: later Institute work applying this paper's attribution logic, including work on measuring relational intelligence, anthropomorphism resistance and multi-centered systems

Scale: Level 1 (dyadic human-AI interaction systems); Level 2 and Level 3 (multi-centered fields and distributed continuity relay) stated in Section 7 as a scaling hypothesis rather than an established extension

Connects to: SF0004 (MTCS-R) and SF0009 as the cross-walk targets for Section 8.2's candidate measurement targets

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